Narrative Weaver: Supporting Reflective Wellbeing in Hybrid Work Through Non-Intrusive AI

Human–Computer Interaction (CSE 4451)
Section B

United International University

Group Loki

Adiba Habiba
Rownok Jahan
Faria Benta Alamgir
Md. Mashrafe Hridoy
Md. Noman Hossain
Safrin Sanzida Islam

Submitted to: Novia Nurain
Department of Computer Science and Engineering

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Abstract

Hybrid and organizational work environments increasingly rely on digital systems, yet employees often experience stress, mental fatigue, and a lack of meaningful reflection on their work. While AI-based tools are commonly introduced to improve productivity, employees report mixed feelings toward such systems, particularly due to concerns around monitoring and autonomy. This project explores how an AI-supported system can support workplace wellbeing through reflection rather than performance evaluation. Through formative research, iterative design, and user evaluation, we present *Narrative Weaver*, a conceptual system designed to offer optional wellbeing reminders and reflective narratives.

1 Introduction & Problem Statement

Moving to hybrid work has changed more than just where we sit; it has changed how we feel about our jobs. Even though companies talk about big missions and goals, it's easy for an individual employee to feel disconnected when they are just staring at a screen at home. We call this the "purpose gap." When people do not see the value of their work, their motivation and mental wellbeing take a hit.

We wanted to see if Generative AI could help fix this—not by doing the work for the employee, but by acting as a "Narrative Weaver" that connects the dots between a user's tasks and the organization's success. However, we knew from the start that workplace AI can feel intrusive or overly corporate if it is not designed carefully. Our research focused on one main question: *How can we design AI-generated narratives that make employees feel valued and supported without crossing ethical lines or feeling intrusive?* [1]

2 Literature Review

Prior HCI and CSCW research highlights that wellbeing at work is influenced by autonomy, clarity of tasks, and a sense of meaning. Reflective tools such as journaling systems and wellbeing prompts have been shown to support emotional awareness when participation is voluntary. Conversely, productivity-driven monitoring systems often increase stress and reduce trust [2].

Research on AI in the workplace shows both promise and risk. While AI can assist decision-making and efficiency, opaque systems raise ethical and emotional concerns among employees. These studies informed our design approach, emphasizing optional interaction, transparency, and reflective support rather than surveillance or performance optimization [1].

3 Methods

We followed a human-centered design process consisting of formative research, design exploration, and evaluation. Formative Research Three semi-structured interviews were conducted with employees from industry and academic settings to explore workload, stress, meaning, and perceptions of AI. A two-day diary study with two participants captured daily reflections on task meaning and emotional state. One participatory design activity explored early ideas around system interaction and boundaries. Analysis Data from interviews, diary entries, and prototype feedback were analyzed using a lightweight affinity diagramming approach. Recurring themes included workload pressure, autonomy, trust in AI, and emotional fatigue.

4 Findings

Participants described heavy workload, unclear requirements, and sudden deadlines as major contributors to stress. Long working hours intensified fatigue, while meaningful outcomes such as successful releases or positive feedback improved emotional wellbeing. Attitudes toward AI varied: some participants were comfortable using AI for reflection, while others emphasized that wellbeing depends on organizational culture and expressed concerns about monitoring. These findings highlight the need for optional, transparent, and non-intrusive wellbeing support

5 System Design

Based on the findings, we designed Narrative Weaver, a conceptual AI-supported system focused on reflection rather than performance tracking. Supported Tasks Start-of-day reflection: Optional input of mood and workload expectations. Midday wellbeing reminders: Gentle prompts for breaks, hydration, or stretching. End-of-day narrative reflection: A short, supportive summary of the day's work.

Low-Fidelity Prototype Low-fidelity workflow prototypes explored interaction timing, tone, and user control.

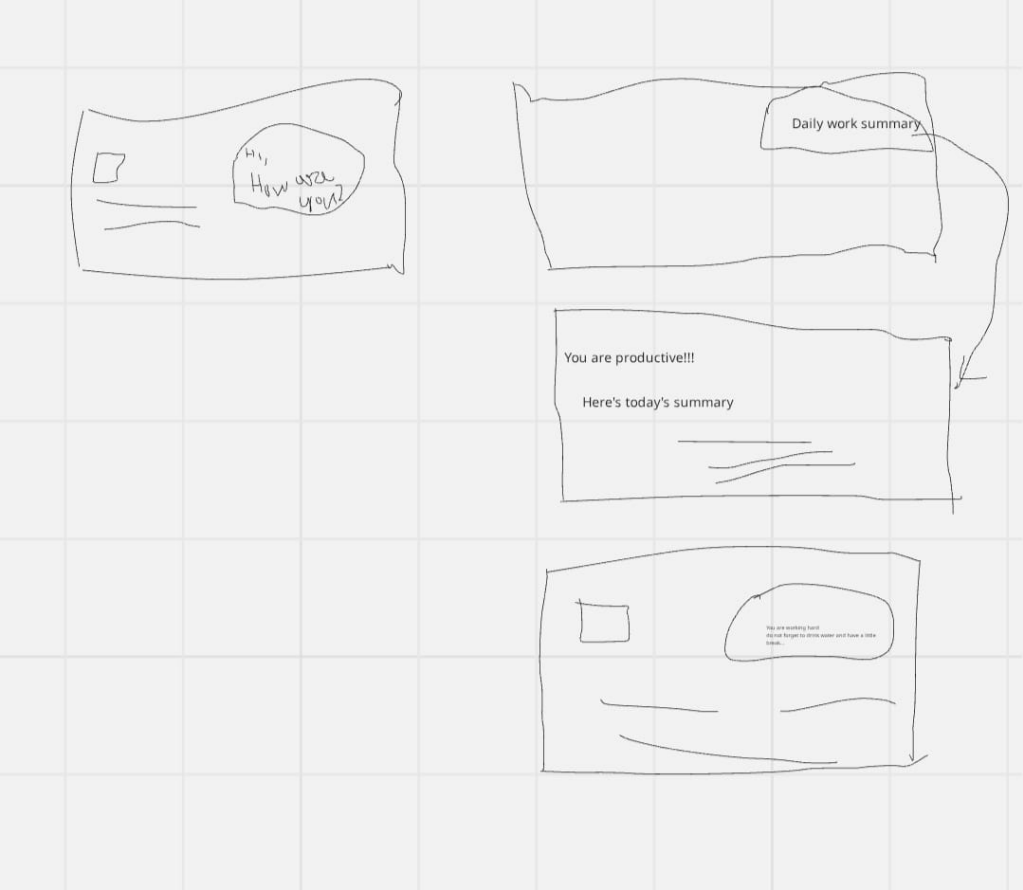


Figure 1: Low-fidelity prototype illustrating optional wellbeing reminders such as stretching, hydration, and short breaks during work hours.

High-Fidelity Prototype The high-fidelity prototype refined layout, language, and interaction flow to reduce intrusiveness and emphasize transparency.

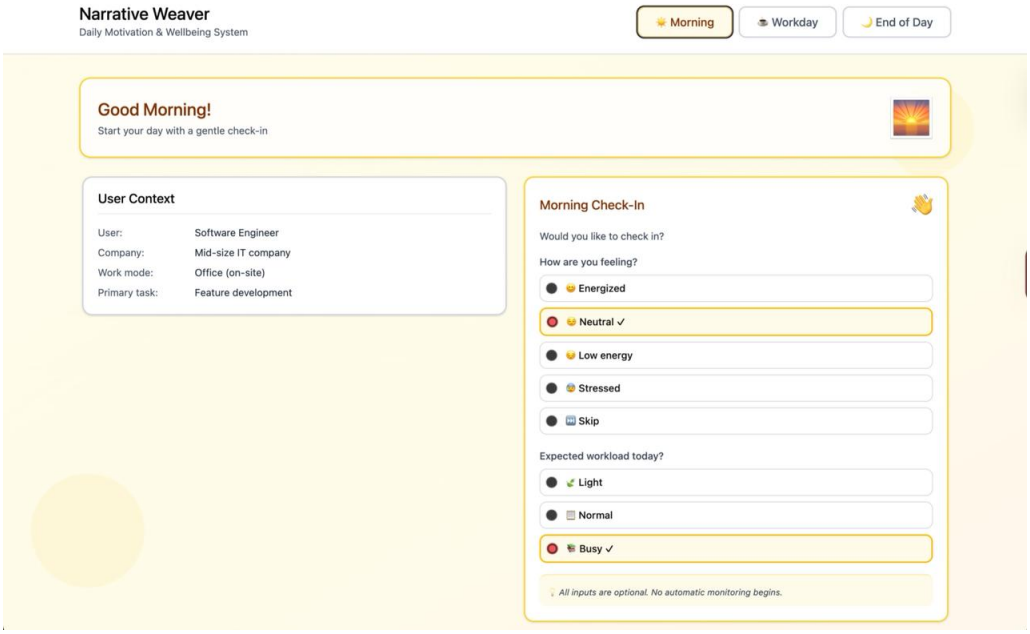


Figure 2: High-fidelity prototype evaluated with users to assess usability and comfort with reflective wellbeing interactions.

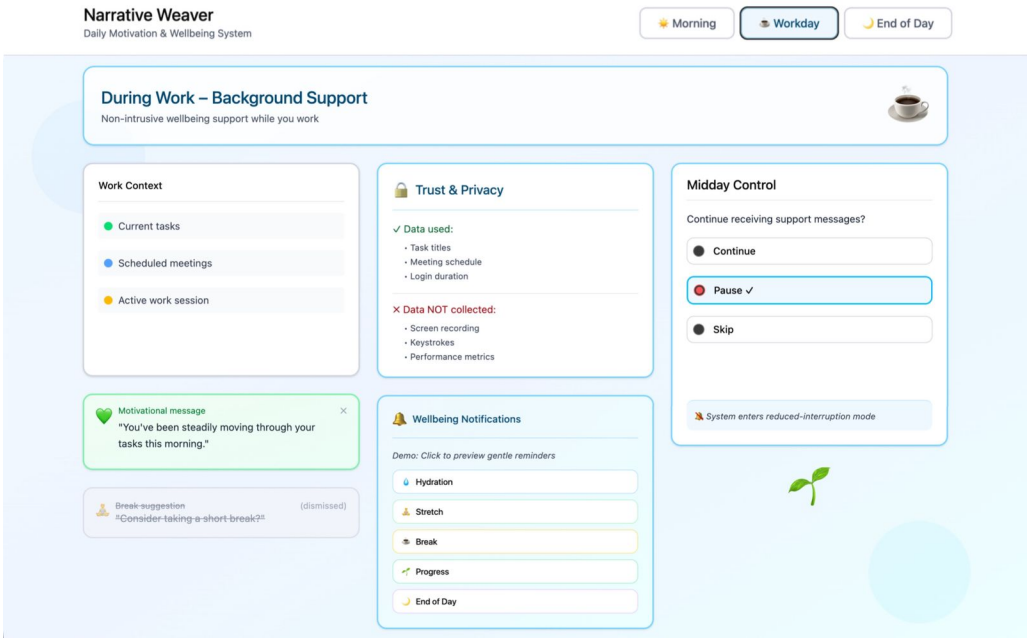


Figure 3: High-fidelity prototype evaluated with users to assess usability and comfort with reflective wellbeing interactions.

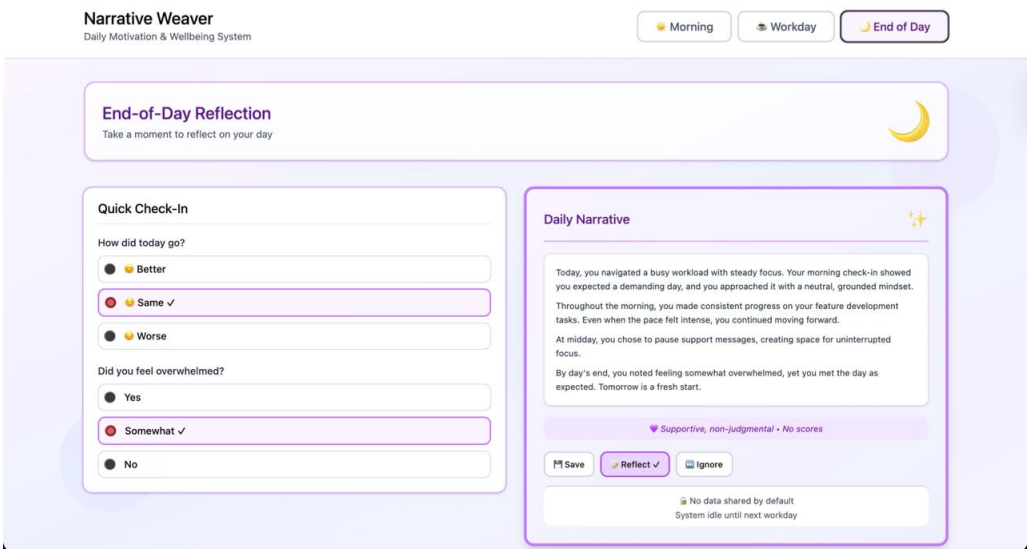


Figure 4: High-fidelity prototype evaluated with users to assess usability and comfort with reflective wellbeing interactions.

6 User Evaluation

A qualitative evaluation was conducted with three participants using semi-structured discussions. Participants reviewed the high-fidelity prototype and shared feedback on usability, comfort, and perceived usefulness. Participants appreciated the optional nature of the reminders and the non-judgmental tone. However, concerns about monitoring persisted, reinforcing the need for clear communication about data use. Feedback suggested that wellbeing support is most acceptable when minimal, infrequent, and clearly separated from productivity metrics.

7 Discussion

Our journey from initial interviews to the final prototype highlighted the delicate balance between technology and wellbeing. Participants were interested in AI systems that acknowledge effort and meaning, but strongly resisted surveillance-oriented designs. Concerns about dignity, fairness, and autonomy influenced our emphasis on transparency and user control [3]. Narrative Weaver works best as a reflective mirror for individual progress rather than a monitoring tool for organizations.

This led to some major shifts in our design. We realized the AI shouldn't act like a manager; it should act like a supportive colleague. That's why we prioritized "Midday Control" and simplified our notifications. In our evaluation, users told us they liked the non-intrusive nature of the wellbeing reminders, but they also reminded us that AI can't replace a healthy company culture. If a workplace is toxic, a supportive AI message might even feel frustratingly fake. This taught us that Narrative Weaver works best as a "reflective mirror" for the user's own progress, rather than a top-down tool

for the company to monitor them. We also learned that the language the AI uses is everything—it needs to be supportive and human, not evaluative or performance-based.

8 Conclusion

The Narrative Weaver project allowed us to explore how HCI can tackle the emotional challenges of the modern workplace. Through our research, design, and testing, we've shown that AI has real potential to improve employee wellbeing by making work feel more meaningful and less like a series of disconnected tasks.

However, our project also serves as a reminder that technology is only one piece of the puzzle. For a tool like Narrative Weaver to truly succeed, it must be built on a foundation of trust and transparency. We found that users are willing to engage with AI for emotional support as long as they feel in control of their data and the interaction. Ultimately, our goal was to create a system that helps people pause and reflect in a fast-paced digital world, and we believe our findings provide a solid roadmap for making workplace AI feel more human, ethical, and genuinely helpful.

References

[1] Ehsan, U., & Riedl, M. (2020). *Human-Centered Explainable AI: Towards a Reflective Sociotechnical Approach*. In *Human-Computer Interaction*. Springer.

[2] Li, I., Dey, A., & Forlizzi, J. (2010). A stage-based model of personal informatics systems. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*.

[3] Binns, R., Van Kleek, M., Veale, M., Lyngs, U., Zhao, J., & Shadbolt, N. (2018). "It's reducing a human being to a percentage": Perceptions of justice in algorithmic decisions. *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*.

A Appendix: Project Resources and Materials

This appendix provides links to additional project materials used during the formative, design, and evaluative stages of the Narrative Weaver project. These resources are included to support transparency and reproducibility of the study process.

A.1 Prototype Design

The high-fidelity digital prototype was designed using Figma and was used during the design and evaluation phases of the project.
Figma Prototype: `https://www.figma.com/make/OKx6ayWrwWqlChfWWZLSA4/Narrative-Weaver-Pro`

A.2 Diary Study Materials

A diary study was conducted using a Google Form to capture daily reflections from participants. Responses were collected and analyzed to understand in-the-moment experiences related to workload, meaning, and wellbeing.
Diary Study Google Form: `https://docs.google.com/forms/d/e/1FAIpQLScWUNWWn_yrbAcd4LJhvwD`
Diary Study Responses (Spreadsheet): `https://docs.google.com/spreadsheets/d/1Tdt17zKkPRRWZ`

A.3 Interview Materials

Semi-structured interviews were conducted with participants to explore experiences related to workplace wellbeing, workload, and perceptions of AI-supported systems. Audio recordings and related materials are stored in the following folder.
Interview Recordings and Notes: `https://drive.google.com/drive/folders/1xpYudNqWvlnBMqkSrW`